

Chemistry

CYL1010



Instructor: Dr. Suresh Sarkar

Email: sarkar@iitj.ac.in

Office: Chemistry Department, #319

Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		
Objectives 1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving 2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs. Learning Outcomes 1. Understanding the behavior of matter and materials using fundamental knowledge of their nature. 2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas. 3. Predict potential complications from combining various chemicals or metals in an engineering setting. Contents <i>Atomic and Molecular Structure [6 Lectures]:</i> particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions <i>Solid State Chemistry [6 Lectures]:</i> Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties. <i>Surface Chemistry[3 Lectures]:</i> Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension. <i>Electrochemical Systems [5 Lectures]:</i> Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion <i>Basic Spectroscopy [7 Lectures]:</i> Selection Rules, Fundamentals FTIR, Raman, UV-visible and Fluorescence spectroscopy <i>Transition Metal Chemistry [5 Lectures]:</i> Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry <i>Organic Chemistry [8 Lectures]:</i> Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies. Textbooks 1. Silberberg, M., <i>Chemistry: The Molecular Nature of Matter and Change</i>, 6th Edition, McGraw Hill Education. 2. West, A.R., (2015), <i>Solid State Chemistry and Its Applications</i>, 2nd edition, John Wiley & Sons. 3. J. D. Lee, <i>Concise Inorganic Chemistry</i>, 5th Edition. 4. Atkins, <i>Physical Chemistry</i>, 11th Edition. Reference Books 1. McMurry, J. E. & Fay, R. C. <i>Chemistry</i>, 5th Edition, Pearson. 2. Jonathan Clayden, Nick Greeves, Stuart Warren, <i>Organic Chemistry</i>, 2nd Edition. 3. F. Albert Cotton, Geoffrey Wilkinson, <i>Basic Inorganic Chemistry</i>, Wiley Student Edition. Self Learning Material 1. https://nptel.ac.in/downloads/122101001/			

Contents

- **Surface Chemistry[3 Lectures]:** Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.
- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
- **Electrochemical Systems [5 Lectures]:** Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion

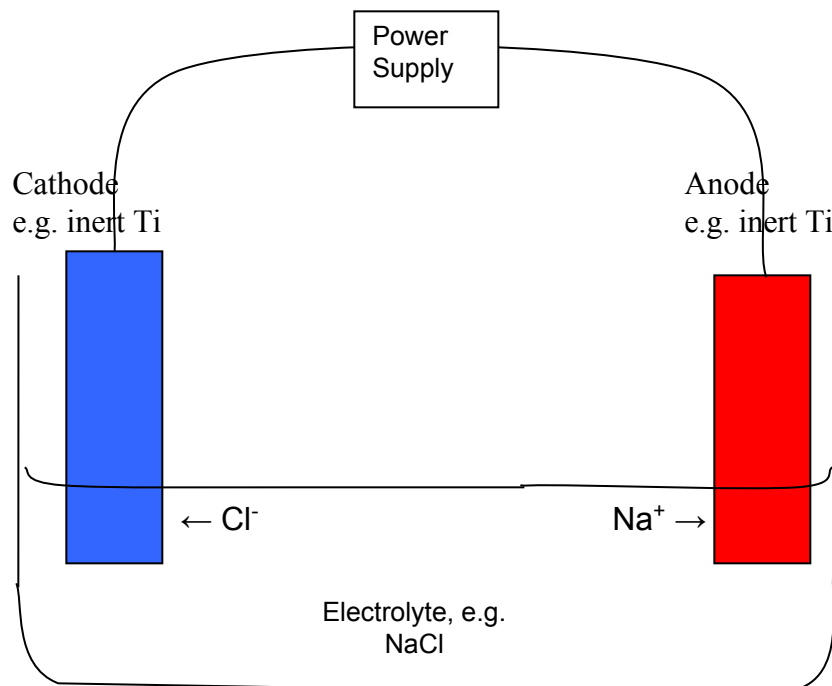


Applications of Cells

Electrolytic cells

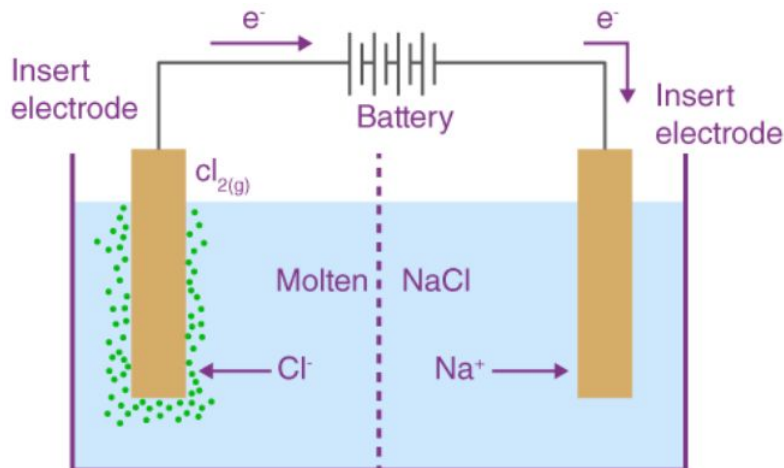
Electrolytic cells

- Use an applied voltage to carry out a nonspontaneous chemical reaction $\Rightarrow \Delta G > 0$
- Electric current supplied by an external source



- External source must provide a greater potential than that for the spontaneous reverse reaction
- **Electrolysis** = process in which electrical energy is used to cause a non-spontaneous chemical reaction to occur

Molten sodium chloride (NaCl) can be subjected to electrolysis with the help of an electrolytic cell, as illustrated below.



Here, two inert electrodes (Pt) are dipped into molten sodium chloride (which contains dissociated Na⁺ cations and Cl⁻ anions). When an electric current is passed into the circuit, the cathode becomes rich in electrons and develops a negative charge. The positively charged sodium cations are now attracted towards the negatively charged cathode. This results in the formation of metallic sodium at the cathode.

Simultaneously, the chlorine atoms are attracted to the positively charged anode. This results in the formation of chlorine gas (Cl₂) at the anode (which is accompanied by the liberation of 2 electrons, finishing the circuit). The associated chemical equations and the overall cell reaction are provided below.

Reaction at Cathode: $[\text{Na}^+ + \text{e}^- \rightarrow \text{Na}] \times 2$

Reaction at Anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

Cell Reaction: $2\text{NaCl} \rightarrow 2\text{Na} + \text{Cl}_2$

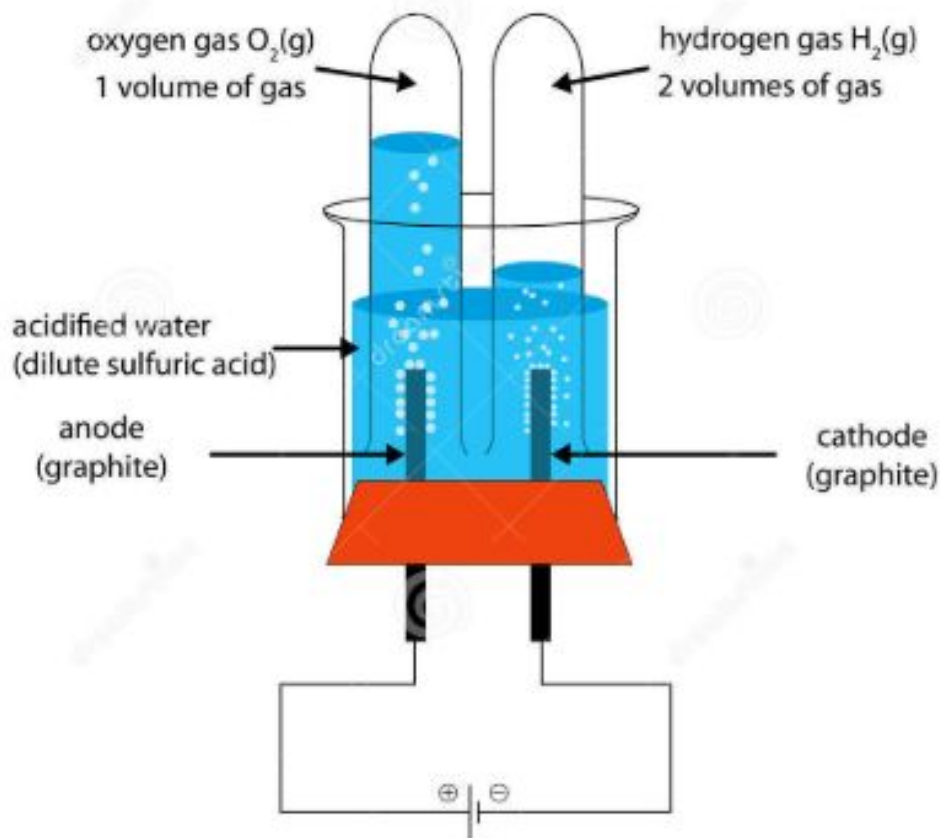
Thus, molten sodium chloride can be subjected to electrolysis in an electrolytic cell to generate metallic sodium and chlorine gas as the products.

Applications of Electrolytic Cells

❖ **Electrolytic cell:** Electrolysis, electroplating....etc.

- ✓ The primary application of electrolytic cells is for the production of oxygen gas and hydrogen gas from water.
- ✓ Another notable application of electrolytic cells is in electroplating, which is the process of forming a thin protective layer of a specific metal on the surface of another metal.
- ✓ They are also used for the extraction of aluminium from bauxite.
- ✓ The electrorefining of many non-ferrous metals is done with the help of electrolytic cells.
- ✓ Such electrochemical cells are also used in electrowinning processes.
- ✓ It can be noted that the industrial production of high-purity copper, high-purity zinc, and high-purity aluminium is almost always done through electrolytic cells.

Electrolysis of water



At the cathode: $4e^- + 4H^+(aq) \rightarrow 2H_2(g)$

At the anode: $2O^{2-}(aq) - 4e^- \rightarrow O_2(g)$

Overall: $2O^{2-}(aq) + 4H^+(aq) \rightarrow 2H_2(g) + O_2(g)$

- Electrolysis of water is the decomposition of water into oxygen and hydrogen gas due to the passage of an electric current.
- The reaction has a standard potential of -1.23 V , meaning it ideally requires a potential difference of 1.23 volts to split water.
- In a non-spontaneous reaction the E_{net}^0 is always less than zero.

Why to split

- Because 70% of the earth is made of water and water is abundantly available.
- Hydrogen fuel production has gained increased attention as oil and other nonrenewable fuels become increasingly depleted and expensive.
- Can be a key component of hydrogen economy.
- By using the process of water splitting, the emission of greenhouse gases (especially Carbon dioxide) can be reduced.

Need For Hydrogen

- Water and solar light are renewable energy resources which on splitting produce hydrogen and oxygen.

Hydrogen is highest energy yield per unit mass among the various fuels: **122kj/g that of fossil fuel is 40kj/g**

- Hydrogen is environment friendly and has zero negative impact on environment
- Major outcome of hydrogen consumption is water which does not harm environment and also contains hydrogen

Current Energy System

- Current energy system is based upon fossil fuels consumption.
- Fossil fuels such as coal, natural gas and oil are consumed to meet the energy demand round the globe.
- Fossil fuels reserves are finite and depleting rapidly because of increase in population and improvement in living standard.
- Other energy resources are wind energy, nuclear energy and solar energy.
- Limitations of cost and efficiency are associated with all of them are responsible for their limited use.

Methodsof watersplitting

Electrolysis

Photoelectrochemical watersplitting

Photocatalytic watersplitting

Radiolysis

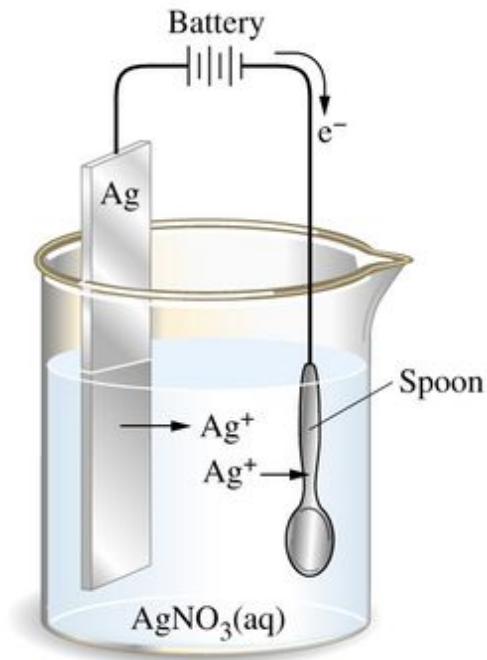
Photobiological watersplitting

Thermal decomposition of water

Nuclear thermal

Chemical production, etc

Electroplating



Electroplating is a procedure that uses electrolysis to apply a thin layer of a metal over the surface of another metal.

Electroplating: Specific case of Ag-electroplating

Purpose: To coat a metal object (substrate) with a thin layer of silver for **decoration, corrosion resistance, or conductivity**.

Apparatus Setup:

- 1. Anode (Positive Electrode):** Pure silver (Ag).
- 2. Cathode (Negative Electrode):** The object to be plated (e.g., copper, nickel).
- 3. Electrolyte:** A solution of **silver nitrate (AgNO_3)**.
- 4. Power Supply:** Direct current (DC source).

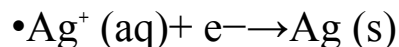
Reaction at Electrodes:

At the Anode (Ag^+ formation):



Silver from the anode dissolves into the solution as Ag^+ ions.

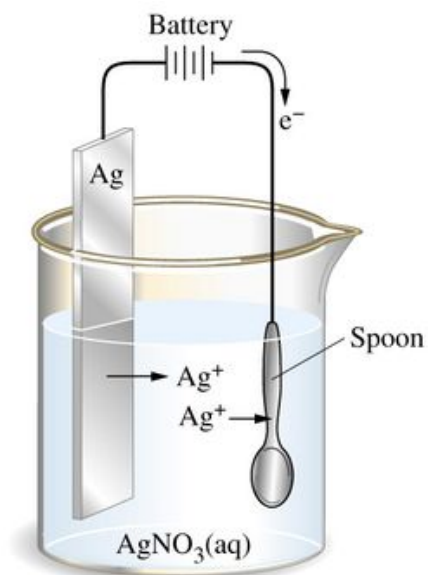
At the Cathode (Ag deposition):



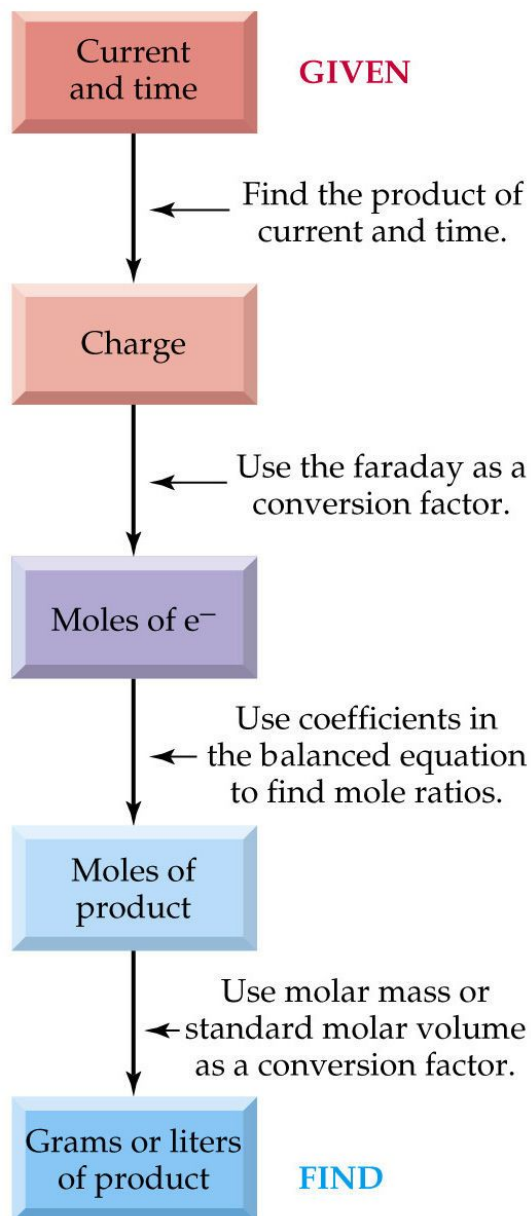
Silver ions are reduced and deposit as a thin, shiny layer on the object.

Steps of Electroplating:

1. Clean the cathode (object to be plated) thoroughly to remove grease, dirt, and oxides.
2. Immerse both cathode and anode into AgNO_3 electrolyte solution.
3. Connect the electrodes to a DC power supply:
 1. Cathode to **negative terminal**
 2. Anode to **positive terminal**
4. Turn on the current; silver ions move from the anode to the cathode and deposit as a uniform layer.
5. After plating, remove the object, rinse with water, and polish if necessary.



Calculating plating



- Have to count charge.
- Measure current I (in amperes)
- 1 amp = 1 coulomb of charge per second
- $q = I \times t$
- $q/nF = \text{moles of metal}$
- Mass of plated metal
- Faraday Constant (F)
(96,480 C/mol e⁻) gives the amount of charge (in coulombs that exist in 1 mole of electrons passing through a circuit.
- 1 volt = 1 joule/coulomb

Calculating plating

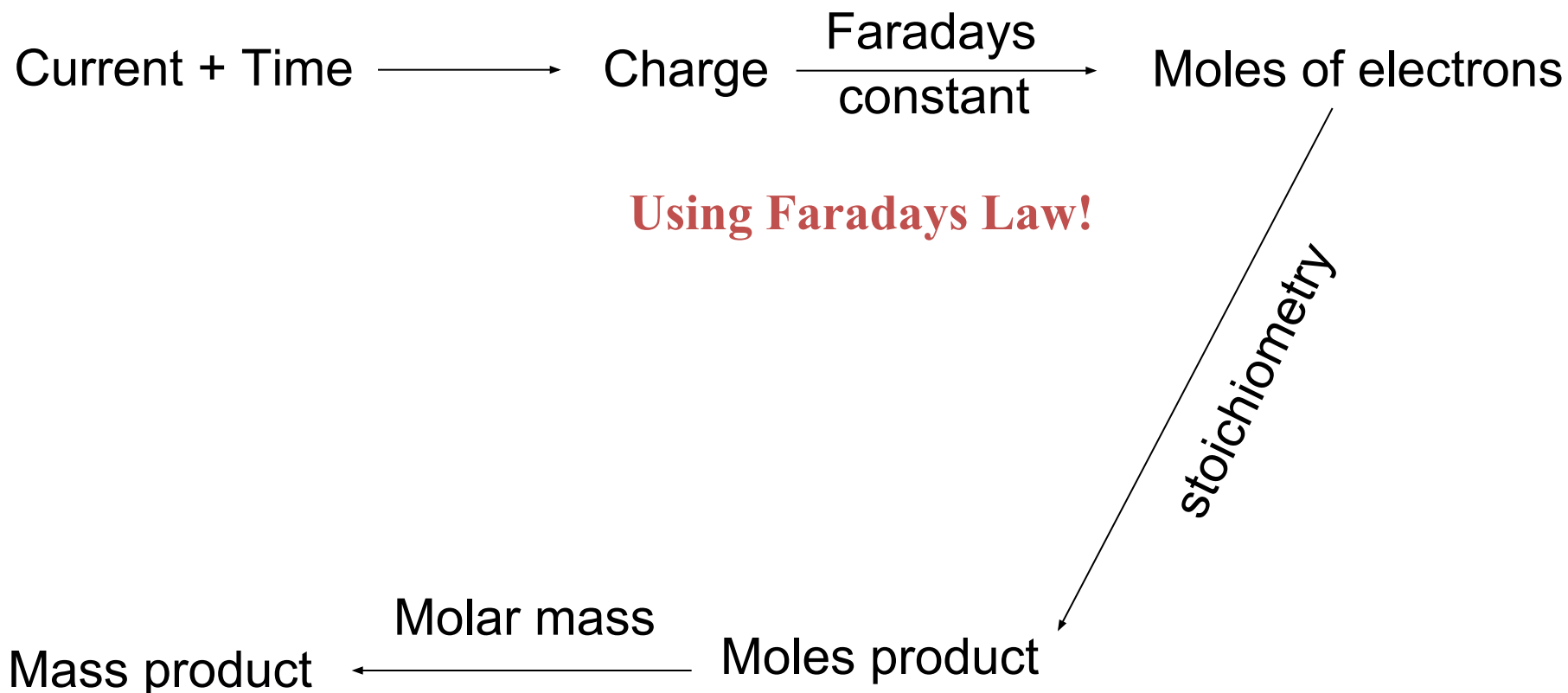
1. Current x time = charge
2. Charge / Faraday = mole of e^-
3. Mol of e^- to mole of element or compound
4. Mole to grams of compound

or the reverse these steps
if you want to find the
time to plate

How many grams of copper are deposited on the cathode of an electrolytic cell if an electric current of 2.00A is run through a solution of CuSO_4 for a period of 20min?

How many hours would it take to produce 75.0g of metallic chromium by the electrolytic reduction of Cr^{3+} with a current of 2.25 A?

Faraday's Law of Electrolysis: the quantity (moles) of product formed by an electric current is stoichiometrically equivalent to the amount (moles) of electrons supplied



Problem: How many grams of copper are deposited on the cathode of an electrolytic cell if an electric current of 2.00A is run through a solution of CuSO_4 for a period of 20min?

How many grams of copper are deposited on the cathode of an electrolytic cell if an electric current of 2.00A is run through a solution of CuSO_4 for a period of 20min?

Answer:



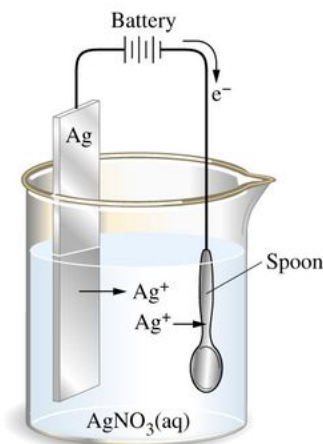
$$2.00\text{A} = 2.00\text{C/s and } 20\text{min (60s/min)} = 1200\text{s}$$

$$\text{Coulombs of e}^- = (2.00\text{C/s})(1200\text{s}) = 2400\text{C}$$

$$\text{mol e}^- = (2400\text{C})(1\text{mol}/96,480\text{C}) = .025\text{mol}$$

$$(.025\text{mol e}^-)(1\text{mol Cu}/2\text{mol e}^-) = .0125\text{mol Cu}$$

$$\text{g Cu} = (.0125\text{mol Cu})(63.55\text{g/mol}) = .79\text{g}$$



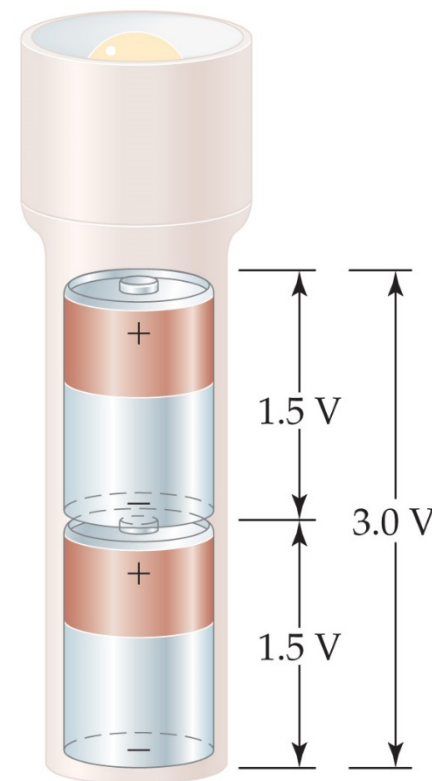
What is a battery?

- It is something that converts chemical energy into electrical energy.
- They produce current via redox reactions:
 - Red=reduction (gaining electrons)
 - Ox=oxidation (losing electrons)
- Batteries are essentially Voltaic Cells connected in series.

a cell or connected group of cells that converts chemical energy into electrical energy by reversible chemical reactions and that may be recharged by passing a current through it in the direction opposite to that of its discharge -- called also storage cell.

❖ **Batteries:** *a portable, self-contained electrochemical power source that consists of one or more voltaic cells.*

- Batteries can be primary cells (cannot be recharged when “dead”—the reaction is complete) or secondary cells (can be recharged).



All primary batteries are electrochemical cells, but they are single-use and not rechargeable.

Electrochemical cell: Batteries

Definition:

- An **electrochemical cell** is a device that converts **chemical energy into electrical energy** through redox (oxidation-reduction) reactions.
- A **primary battery** is a type of electrochemical cell that is **designed to be used once and cannot be recharged** because the chemical reactions are not easily reversible.

General Working Principle:

1. **Anode (negative electrode):** Undergoes **oxidation** (loses electrons).
2. **Cathode (positive electrode):** Undergoes **reduction** (gains electrons).
3. **Electrolyte:** Provides ions to maintain charge balance.
4. **External circuit:** Electrons flow from anode → cathode to do electrical work.

Examples of Primary Batteries:

- **Alkaline battery (Zn/MnO₂)**
- **Lithium primary battery (Li/FeS₂)**

Types of Batteries

The **primary battery** converts chemical energy to electrical energy **directly**, using the chemical materials within the cell to start the action.

The **secondary battery** must first be **charged** with electrical energy before it can convert chemical energy to electrical energy.

The **secondary battery** is frequently called a **storage battery**, since it stores the energy that is supplied to it.

Definition of Cell:

- A primary cell cannot be recharged because the internal chemical reaction cannot be restored.

Example: ZINC CARBON (1.5V), ALKALINE (1.5V)



- A secondary cell, or storage cell, can be recharged because its chemical reaction is reversible. Example: LEAD ACID (2.0V), NICKEL - CADMIUM (1.2V), NICKEL - METAL HYDRIDE (1.2V), LITHIUM – ION (3.3V)



Primary cells

Battery Type	Chemical Composition	Recharge State	Discharge State	Electrolyte Used	Common Uses
Alkaline	Zn / MnO ₂	Anode: Zn (metal) Cathode: MnO ₂	Anode: Zn(OH) ₂ Cathode: Mn ₂ O ₃ or MnOOH	Potassium hydroxide (KOH)	Remotes, clocks, toys
Zinc-Carbon	Zn / MnO ₂	Anode: Zn (metal) Cathode: MnO ₂	Anode: Zn ²⁺ salts Cathode: Mn ₂ O ₃ or MnOOH	Ammonium chloride (NH ₄ Cl) + Zinc chloride	Flashlights, toys
Zinc-Chloride	Zn / MnO ₂	Anode: Zn Cathode: MnO ₂	Anode: Zn ²⁺ salts Cathode: MnOOH	Zinc chloride (ZnCl ₂)	Higher-drain flashlights, toys
Lithium (MnO ₂)	Li / MnO ₂	Anode: Li (metal) Cathode: MnO ₂	Anode: Li ⁺ Cathode: LiMnO ₂ or MnOOH	Lithium salt in organic solvent (e.g., LiCF ₃ SO ₃)	Cameras, high-drain devices
Lithium (SOCl ₂)	Li / Thionyl chloride (SOCl ₂)	Anode: Li Cathode: SOCl ₂	Anode: LiCl Cathode: SO ₂ , S, LiAlCl ₄	Thionyl chloride (acts as electrolyte too)	Memory backup, military, alarms
Silver-Oxide	Zn / Ag ₂ O	Anode: Zn Cathode: Ag ₂ O	Anode: Zn(OH) ₂ Cathode: Ag (metal)	Potassium hydroxide (KOH) or Sodium hydroxide (NaOH)	Watches, calculators
Mercury (Obsolete)	Zn / HgO	Anode: Zn Cathode: HgO	Anode: ZnO Cathode: Hg (metal)	Potassium hydroxide (KOH)	(Obsolete) watches, hearing aids
Zinc-Air	Zn / O ₂ (from air)	Anode: Zn Cathode: O ₂	Anode: ZnO Cathode: OH ⁻	Potassium hydroxide (KOH)	Hearing aids, medical devices
Magnesium	Mg / CuCl ₂ or similar	Anode: Mg Cathode: CuCl ₂	Anode: Mg ²⁺ Cathode: Cu (metal)	Salt water or MgCl ₂ solution	Military, emergency applications
Thermal	Varies (e.g., Li-alloy / FeS ₂)	Anode: Li alloy Cathode: FeS ₂	Anode: Li ⁺ Cathode: Fe, Li ₂ S	Molten salts (e.g., LiCl-KCl eutectic)	Missiles, space, defense
Reserve	Varies; electrolyte added at activation	Depends on type (dry until activated)	Same as applicable battery chemistry	Typically acid or base, released on demand	Missiles, emergency systems

Primary batteries play a crucial role in modern electronics, with each type optimized for specific performance and cost targets.

- **Primary cells cannot be recharged** safely — trying to recharge them can cause **leakage, fire, or explosion**.
- **Lithium primary** cells are different from **Lithium-ion** (which are rechargeable).
- **Mercury batteries are largely phased out due** to environmental concerns.
- **Thermal and reserve batteries** are niche and mainly used in defense and aerospace.

Most Popular / Common Primary Batteries

Battery Type	Typical Applications	Why Popular?
Alkaline	AA/AAA batteries, toys, remotes, flashlights	Inexpensive, long shelf life, widely used
Zinc-Carbon	Flashlights, remotes (low-drain devices)	Very cheap, common in developing markets
Zinc-Chloride	Slightly better than zinc-carbon	Better capacity, used where alkaline is not needed
Lithium (MnO₂)	Cameras, high-drain devices	Lightweight, high energy, long shelf life
Silver-Oxide	Watches, calculators, hearing aids	Small size, stable voltage
Zinc-Air	Hearing aids, medical devices	High energy-to-weight ratio

Battery Type	Voltage (Nominal)	Energy Density	Shelf Life	Cost	Use Case
Alkaline	1.5V	Medium	5–10 years	Low	General household
Zinc-Carbon	1.5V	Low	2–3 years	Very Low	Toys, remotes
Zinc-Chloride	1.5V	Medium	3–5 years	Low	Flashlights, radios
Lithium (MnO ₂)	3.0V	High	10–15 years	Medium	Cameras, key fobs
Silver-Oxide	1.55V	High (small size)	4–6 years	High	Watches, calculators
Zinc-Air	1.4V	Very High	~3 years (sealed)	Medium	Hearing aids, medical

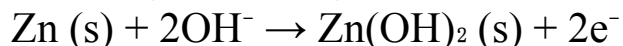
- **Alkaline:** Best all-rounder for general use.
 - **Lithium (MnO₂):** For high-drain and long-life applications.
 - **Silver-Oxide & Zinc-Air:** Best for compact, low-power devices like watches and hearing aids.
 - **Zinc-Carbon/Zinc-Chloride:** Budget options, good for low-drain use only.
- Each has trade-offs between **cost**, **energy density**, **shelf life**, and **application**, making it essential to match the right battery to your specific need.

Alkaline Battery

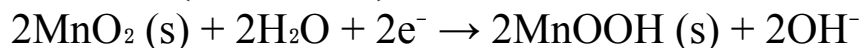
Working Principle

Alkaline batteries generate electricity through a **redox reaction** between **zinc (Zn)** (anode) and **manganese dioxide (MnO₂)** (cathode), in the presence of an **alkaline electrolyte**, typically **potassium hydroxide (KOH)**.

- **Anode (Oxidation):**

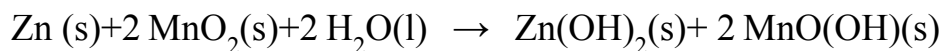


- **Cathode (Reduction):**



The electrons flow through the external circuit to power the device.

Net cell reaction:



Advantages

- Widely available and inexpensive
- Long shelf life (5–10 years)
- Higher energy density than zinc-carbon
- Non-toxic (compared to mercury cells)

Disadvantages

- Not suitable for high-drain applications over time
- Can leak over long periods
- Voltage drops over discharge (not constant)



Alkaline battery

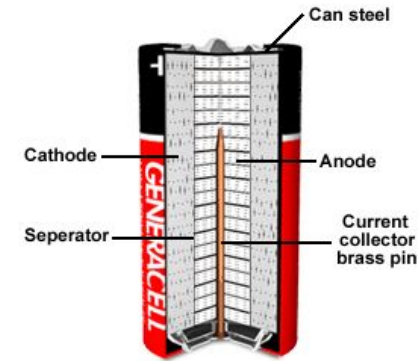
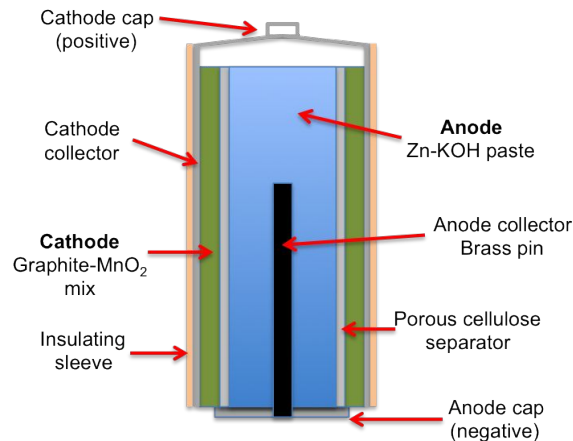
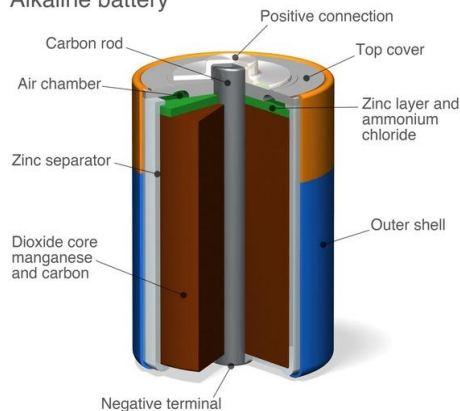


Figure 1

Main Parts in the Diagram

1. Positive connection (Top terminal)

- This is the **cathode (+ terminal)** of the battery.
- Connects to the **manganese dioxide (MnO₂) cathode material**.

2. Carbon rod

- Acts as a conductor, allowing electrons to flow from the chemical reaction to the external circuit.

3. Air chamber

- Provides space for gases that may form inside during chemical reactions.

4. Zinc layer and ammonium chloride

- The **zinc** is the **anode (- terminal)** material.
- Ammonium chloride (or KOH in modern alkaline cells) is the **electrolyte** that enables ion movement.

5. Zinc separator

- A porous barrier that prevents the zinc anode from directly touching the manganese dioxide cathode (which would cause a short circuit).
- Allows ions to pass through.

6. Dioxide core (manganese dioxide + carbon)

- The **cathode material** (MnO₂ mixed with carbon for conductivity).
- Manganese dioxide undergoes reduction during discharge.

7. Outer shell

- Provides structural support and protection.
- Often made of steel.

8. Negative terminal (Bottom)

- The flat end of the battery.
- Connected to the zinc anode.

Role of Carbon (Graphite rod)

- The **carbon (graphite) rod** is **not chemically reactive**.
- Its job is purely **electrical**:
 - Provides a **conductive path** for electrons from MnO₂ to the external circuit.
 - MnO₂ alone is a poor conductor, so the **carbon rod collects current**.

How Electricity Flows

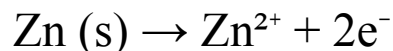
1. Zinc (- terminal) loses electrons (oxidation).
2. Electrons travel through the **external circuit** (powering your device).
3. They enter the carbon rod and reduce MnO₂ at the cathode (+ terminal).
4. The electrolyte allows ionic balance so the reaction continues.

Zinc-Carbon Battery

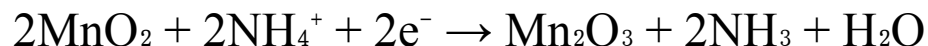
Working Principle

Zinc acts as the anode, while manganese dioxide acts as the cathode. The electrolyte is **ammonium chloride (NH₄Cl)** and/or **zinc chloride (ZnCl₂)** in a paste form.

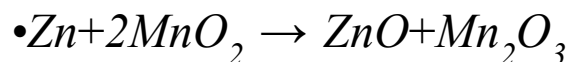
- **Anode (Oxidation):**



- **Cathode (Reduction):**



1. Overall reaction:



Advantages

- Very inexpensive
- Readily available
- Works well in low-drain devices

Disadvantages

- Shorter shelf life
- Low energy density
- Voltage drops quickly
- **Can leak and corrode**

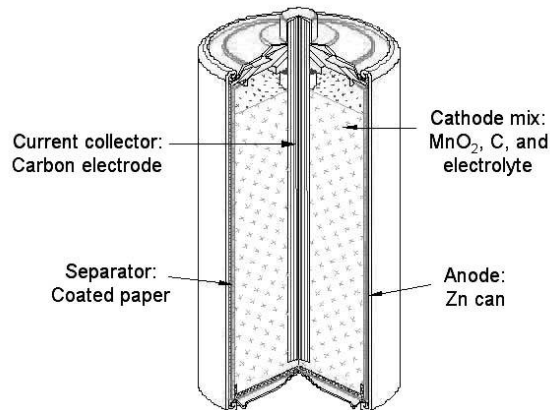
(A)



(B)



Carbon Zinc Cell and Reactions



Electricity Flow

- **Electrons:** From **zinc (-)** → through your device → to **manganese dioxide (+)**.
- **Ions:** Move inside the battery through the electrolyte to balance the charge.

2. Two main parts inside:

- **Zinc (Anode, - terminal at the bottom)** → gives away electrons.
- **Manganese dioxide (Cathode, + terminal at the top)** → accepts electrons.

3. Electrolyte (Ammonium chloride or Potassium hydroxide)

- Works like a “bridge” for ions to move between zinc and manganese dioxide.
- This keeps the reaction going.

4. When battery is connected in a device:

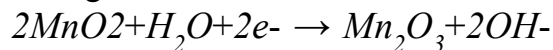
- **Zinc is oxidized (loses electrons):**



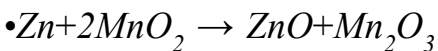
- Electrons flow **out of the negative terminal**, through the device (powering it), and into the **positive terminal**.

5. At the cathode (manganese dioxide + carbon):

- Manganese dioxide **takes those electrons** and gets reduced:



6. Overall reaction:



Why corrosion happens in the casing

In a **Zinc–Carbon battery**, the **zinc can corrode** because:

1. Zinc acts as both the container and the anode

- The zinc casing is in direct contact with the electrolyte paste (NH_4Cl or ZnCl_2).
- Since it is the anode, it naturally undergoes **oxidation**:
- $\text{Zn (s)} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
- This slowly corrodes the zinc.

2. Electrolyte is acidic (NH_4Cl or ZnCl_2)

- The slightly acidic ammonium chloride paste reacts chemically with zinc even when the battery is not in use (“self-discharge”):
- $\text{Zn (s)} + 2 \text{NH}_4^+ \rightarrow [\text{Zn}(\text{NH}_3)_2]^{2+} + \text{H}_2(\text{g})$
- This uncontrolled reaction also corrodes zinc and sometimes causes **leakage** of electrolyte.

3. Hydrogen evolution

1. During discharge or side reactions, hydrogen gas can form at the zinc surface.
2. This creates localized attack (pitting corrosion) and weakens the zinc casing.

Zinc–Carbon battery and Alkaline battery look similar from outside, but they are not the same. They are two different “dry cell” technologies.

Alkaline Battery

- **Anode:** Zinc powder (oxidized to ZnO)
- **Cathode:** Manganese dioxide (MnO_2 , reduced)
- **Electrolyte:** Potassium hydroxide (KOH , alkaline) instead of NH_4Cl
- **Carbon rod:** Not needed (MnO_2 is mixed with conductive additives)
- **Voltage:** ~ 1.5 V per cell (same nominal voltage)
- **Capacity:** Much higher, long-lasting, more stable voltage
- **Cost:** More expensive than zinc–carbon
- **Uses:** Cameras, toys, wireless devices, anything with higher power demand

Zinc–Carbon Battery

- Anode:** Zinc can (oxidized to Zn^{2+})
- **Cathode:** Manganese dioxide (MnO_2 , reduced)
- **Electrolyte:** Ammonium chloride (NH_4Cl) or Zinc chloride (ZnCl_2) in paste form
- **Carbon rod:** Only a conductor, not reacting
- **Voltage:** ~ 1.5 V per cell
- **Capacity:** Low, discharges quickly
- **Cost:** Very cheap
- **Uses:** Remote controls, wall clocks, torches (low-drain devices)

Both use **Zinc (anode)** and **MnO_2 (cathode)** as the actual reactants.

But the **electrolyte is different**, and that’s why **alkaline batteries last much longer**.

Key Difference

- **Electrolyte type:**
 - Zinc–Carbon $\rightarrow$ acidic/neutral salts (NH_4Cl)
 - Alkaline $\rightarrow$ **alkaline KOH**
- **Performance:**
 - Zinc–Carbon = weak, cheap, for light use
 - Alkaline = stronger, longer-lasting, for heavy use

What is the main problem may cause upon using NH_4Cl in Zinc-Carbon battery

Zinc-carbon battery, In an NH_4Cl -only cell, Zn^{2+} can form $\text{Zn}(\text{OH})_2$ (insoluble) due to local pH changes. $\text{Zn}(\text{OH})_2$ increase the cell internal resistance and drops the current.

The Core Problem: A Local "Hot Spot" of High pH

Even though the overall electrolyte (NH_4Cl) is mildly acidic, the chemical reactions at the electrodes create localized conditions where the pH can rise dramatically, leading to the precipitation of $\text{Zn}(\text{OH})_2$.

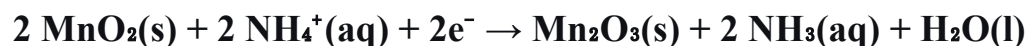
Here is the sequence of events:

Step 1: The Main Discharge Reaction at the Anode (Zinc Can)

At the zinc anode, oxidation occurs:

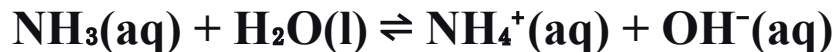
$\text{Zn}(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$, This is the primary reaction that releases electrons for your device.

Step 2: The Main Discharge Reaction at the Cathode (MnO_2 & Carbon Rod): At the cathode, manganese dioxide is reduced. A primary reaction involves the reduction of MnO_2 itself, but a critical side reaction also occurs, especially at lower discharge rates:



This is the key reaction! Notice the product: **Ammonia (NH_3)**

Step 3: Ammonia Creates a Local Basic Environment: Ammonia (NH₃) is a weak base. It reacts with water in the electrolyte:



This reaction generates **hydroxide ions (OH⁻)**, which **increase the pH locally** around the cathode and the separator

Step 4: The "Fatal" Encounter and Precipitation

Now, we have the two products meeting:

- **Zn²⁺ ions** are migrating away from the zinc anode.
- **OH⁻ ions** are being generated near the cathode.

When they meet in the middle (often within the porous separator or near the cathode), they form an insoluble precipitate:



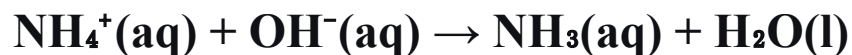
This white, gelatinous solid is highly insoluble and forms a physical barrier.

The Consequences of Zn(OH)_2 Formation

This precipitation of $\text{Zn(OH)}_2(\text{s})$ causes several major problems for the battery:

- 1. Increased Internal Resistance:** The Zn(OH)_2 precipitate clogs the pores of the separator and the cathode. This physically blocks the movement of ions, dramatically increasing the battery's internal resistance.
- 2. Capacity Loss:** The zinc ions that get "locked up" as Zn(OH)_2 are no longer available to participate in the useful current-generating reaction ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$). This effectively reduces the battery's capacity.
- 3. Poor Performance under Load:** As the internal resistance rises, the battery's voltage drops severely under any significant load (e.g., when you try to power a bright flashlight). This is a hallmark of zinc-carbon battery behavior.
- 4. Inefficient Discharge:** A significant amount of the zinc is chemically "wasted" without producing useful electrical current.

How the Battery-Design Tries to Mitigate This issue of Zn(OH)_2 Formation : The abundant NH_4^+ ions in the electrolyte can react with the OH^- ions, acting as a buffer:



This reaction "mops up" some of the hydroxide ions before they can react with Zn^{2+} , helping to keep the local pH lower and preventing some precipitation. However, this buffering capacity is limited and gets overwhelmed, especially as the battery discharges and the NH_4^+ concentration drops

What is the advantage of using NH_4Cl in Zinc-Carbon battery

1. Cost and Manufacturing Simplicity

- Raw Materials:** Ammonium chloride (NH_4Cl) is an abundant, inexpensive chemical. The entire Zinc-Carbon system was designed to be a low-cost, mass-producible power source.
- Construction:** The cell can be constructed with simpler, less refined materials. The separator can be a simple paper or paste barrier, which is sufficient for the less corrosive NH_4Cl electrolyte.

2. Lower Self-Discharge Rate (at room temperature): The chemical reaction in a Zinc-Carbon cell proceeds more slowly than in an alkaline cell. This means that when sitting on a shelf, a Zinc-Carbon battery loses its charge more slowly than an alkaline battery at room temperature.

- Why?** The corrosion reaction between the acidic electrolyte and the zinc anode is less aggressive than the reaction between the basic KOH and the zinc in an alkaline battery. (However, modern alkaline batteries have improved greatly in this area).

3. Historical Precedence and Proven Technology: The Zinc-Carbon (Leclanché) cell is one of the earliest successful commercial battery designs (dating back to the 1860s). The technology was well-understood, and massive manufacturing infrastructure was built around it long before alkaline batteries became commercially dominant in the later 20th century.

What is the advantage of using KOH in Alkaline battery over NH_4Cl in Zinc-Carbon battery

The advantages of using **KOH (a strong base)** in an alkaline battery over **NH_4Cl (a mild acid)** in a zinc-carbon battery are significant and revolve around performance, efficiency, and longevity

1. Superior High-Drain Performance

- KOH:** It is a **strong base and an excellent ionic conductor**. This allows ions to flow through the electrolyte with very low resistance. As a result, alkaline batteries can deliver very high currents (amps) without a significant drop in voltage. This makes them ideal for power-hungry devices like digital cameras, powerful flashlights, motorized toys, and portable audio equipment.
- **NH_4Cl :** The electrolyte has a much **higher internal resistance**. When you try to pull a high current from a zinc-carbon battery, the voltage "sags" or drops dramatically, causing the device to function poorly or shut off.

2. Significantly Higher Energy Density

- KOH:** The electrochemical reactions in an alkaline cell are more efficient and allow for a denser, more active cathode material (manganese dioxide). This means an alkaline battery of the same size (e.g., an AA) can store **up to 5-8 times more electrical energy than a zinc-carbon battery**.
- Result:** Alkaline batteries last much, much longer in all devices, especially in high-drain applications

3. Excellent Low-Temperature Performance

- KOH:** The ionic conductivity of the potassium hydroxide electrolyte remains relatively high even at low temperatures.
- NH₄Cl:** The conductivity drops drastically in the cold. A zinc-carbon battery becomes almost useless in freezing temperatures, while an alkaline battery will still function, albeit with reduced capacity

4. Much Longer Shelf Life

- KOH:** Modern alkaline batteries have a shelf life of **5-10 years** because the corrosion of the zinc anode is well-controlled with special inhibitors and the sealed construction prevents leakage and drying out.
- NH₄Cl:** Zinc-carbon batteries are more prone to leakage and have a shorter shelf life (2-3 years). The acidic electrolyte can slowly react with the zinc can, and water can evaporate more easily, causing the battery to dry out

5. More Stable Voltage Output

- KOH:** Alkaline batteries maintain a relatively stable voltage (around 1.2V - 1.3V under load) for most of their discharge life, providing consistent power to the device.
- NH₄Cl:** Zinc-carbon batteries have a sloping discharge curve. Their voltage starts at 1.5V and steadily declines throughout use, leading to dimming lights or slowing devices.

6. Superior Leakage Resistance

- KOH:** While all batteries can leak, modern alkaline batteries are manufactured with superior seals and anti-corrosion additives that make them far less prone to leaking and damaging devices, especially when stored properly.
- NH₄Cl:** Zinc-carbon batteries are notoriously prone to leaking potassium chloride salts as they age or discharge fully, which can severely damage the electronic device they are left in.

Zinc–Carbon battery and Alkaline battery look similar from outside, but they are not the same. They are two different “dry cell” technologies.

Feature	Zinc–Carbon Battery	Alkaline Battery
Anode	Zinc casing (outer can)	Powdered zinc inside as paste
Cathode	MnO ₂ + C mixture (with central carbon rod)	MnO ₂ packed against steel can (can = cathode collector)
Electrolyte	NH ₄ Cl (slightly acidic)	KOH solution (alkaline, less corrosive to zinc)
Container	Zinc metal (sacrificial → corrodes)	Steel can (inert, strong)
Corrosion issue	Severe — Zn casing attacked even when idle	Much less — Zn powder protected, steel can doesn’t corrode

Zinc-Chloride Battery

Working Principle

Similar to zinc-carbon but uses **zinc chloride** as the main electrolyte for improved capacity and performance.

- Enhanced cathode composition for higher drain
- Electrochemical reactions are nearly the same as in zinc-carbon cells

Advantages

- Higher capacity than standard zinc-carbon
- Longer shelf life
- Improved performance in moderate-drain devices

Disadvantages

- Still inferior to alkaline in terms of capacity and leakage resistance

Not ideal for high-drain applications

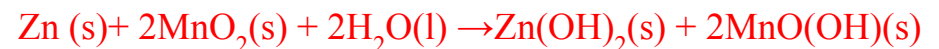


Zinc-Chloride Battery (Heavy-Duty Dry Cell)

Construction

- **Anode (– terminal):** Zinc can (gets oxidized)
- **Cathode (+ terminal):** Manganese dioxide (MnO_2 , gets reduced) mixed with carbon for conductivity
- **Electrolyte:** Paste of **zinc chloride (ZnCl_2)**, sometimes mixed with a little ammonium chloride (NH_4Cl)
- **Carbon rod:** Inert conductor for cathode reactions

Overall Cell Reaction:



Why ZnCl_2 is used instead of NH_4Cl only?

- NH_4Cl -based zinc–carbon cells “dry out” and release ammonia gas (NH_3).
- Replacing most of the NH_4Cl with ZnCl_2 :
 - Reduces gas formation
 - Keeps paste moist longer
 - Improves cell’s life and current capacity

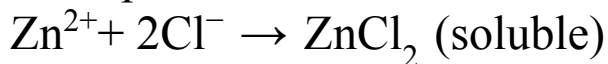
In an NH_4Cl -only cell, Zn^{2+} can form **$\text{Zn}(\text{OH})_2$ (insoluble)** due to local pH changes → this creates passivation and reduces current flow

Role of ZnCl_2 Electrolyte

Provides plenty of Cl^- ions

- ZnCl_2 is highly soluble, dissociating as:
- $\text{ZnCl}_2 \rightleftharpoons \text{Zn}^{2+} + 2\text{Cl}^-$
- The extra **Cl^- ions** immediately combine with the Zn^{2+} formed at the anode due to the oxidation of Zn, preventing precipitation.

Example:



Formation of Complex Ions

- Zn^{2+} doesn’t just exist as free ions — in concentrated chloride solution it forms **chloro-complexes**:
- $\text{Zn}^{2+} + 4\text{Cl}^- \rightarrow [\text{ZnCl}_4]^{2-}$
- These complexes are **highly soluble and stable**, so zinc never precipitates as insoluble hydroxides or oxides.

Why ZnCl₂ instead of NH₄Cl only?

- NH₄Cl-based zinc–carbon cells “dry out” and release ammonia gas (NH₃).
- Replacing most of the NH₄Cl with ZnCl₂:
 - Reduces gas formation
 - Keeps paste moist longer
 - Improves cell’s life and current capacity

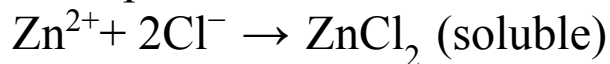
In an NH₄Cl-only cell, Zn²⁺ can form **Zn(OH)₂ (insoluble)** due to local pH changes → this creates passivation and reduces current flow

Role of ZnCl₂ Electrolyte

Provides plenty of Cl[−] ions

- ZnCl₂ is highly soluble, dissociating as:
- $\text{ZnCl}_2 \rightleftharpoons \text{Zn}^{2+} + 2\text{Cl}^-$
- The extra **Cl[−] ions** immediately combine with the Zn²⁺ formed at the anode, preventing precipitation.

Example:



Formation of Complex Ions

- Zn²⁺ doesn’t just exist as free ions — in concentrated chloride solution it forms **chloro-complexes**:
- $\text{Zn}^{2+} + 4\text{Cl}^- \rightarrow [\text{ZnCl}_4]^{2-}$
- These complexes are **highly soluble and stable**, so zinc never precipitates as insoluble hydroxides or oxides.